

Lab No: 01

Course: Data Visualization

Tool: Microsoft Power BI

Lab Duration: 1.5 Hours

Learning Objectives

After completing this lab, students will be able to:

- Import multiple Excel files into Power BI
 - Perform basic data cleaning using **Power Query**
 - Create table **relationships**
 - Use **DAX** to create calculated columns and measures
 - Use built-in functions: **CONCAT / CONCATENATE, SUM, COUNT**
 - Build a basic multi-page Power BI report
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Lab Scenario

You are a **Data Analyst** working for a hospital. The hospital management wants a basic Power BI report to analyze:

- Patients information
- Hospital visits
- Departments and doctors
- Billing summary

You are given **three Excel files** containing hospital data. Your task is to import, clean, model, analyze, and visualize this data using Power BI.

Files Provided

You are provided with the following Excel files:

1. **Patients.xlsx**
2. **Visits.xlsx**
3. **Hospital_Master.xlsx**
 - Contains **Departments** and **Doctors** tables

 Do NOT modify Excel files directly.

Part 1 — Import Data

1. Open **Power BI Desktop**
 2. Click **Get Data** → **Excel**
 3. Import all provided Excel files
 4. Select all relevant tables
 5. Click **Transform Data**
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Part 2 — Data Cleaning using Power Query

A) Patients Table

Perform the following steps:

- Ensure **DOB** column data type is **Date**
- Ensure **PatientID** is **Text**
- Trim spaces from:
 - **FirstName**
 - **LastName**
- Check for null or blank values

B) Visits Table

Perform the following steps:

- Set **VisitDate** data type to **Date**
- Set **BillAmount** to **Decimal Number**
- Filter records where:
 - $\text{BillAmount} \leq 0$ (remove invalid entries)
- Ensure all ID columns are **Text**

C) Doctors Table

- Trim **DoctorName**
- Ensure **DepartmentID** data type matches Departments table

After completing all steps:

 Click **Close & Apply**

Part 3 — Create Relationships

Switch to **Model View** and create the following relationships:

1. **Patients[PatientID] → Visits[PatientID]**
2. **Doctors[DoctorID] → Visits[DoctorID]**
3. **Departments[DepartmentID] → Visits[DepartmentID]**
4. **Departments[DepartmentID] → Doctors[DepartmentID]**

Relationship settings:

- Cardinality: **One-to-Many**
 - Cross-filter direction: **Single**
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Part 4 — DAX Calculations

A) Full Name (Calculated Column)

Create a new calculated column in **Patients** table:

```
Full Name = Patients[FirstName] & " " & Patients[LastName]
```

(Alternate solution)

```
Full Name = CONCATENATE(Patients[FirstName], " " & Patients[LastName])
```

B) Age Calculation from DOB

Create a calculated column in **Patients** table:

```
Age = DATEDIFF(Patients[DOB], TODAY(), YEAR)
```

C) Measures (Built-in Functions)

Create the following measures:

Total Billing

```
Total Billing = SUM(Visits[BillAmount])
```

Total Visits

```
Total Visits = COUNT(Visits[VisitID])
```

Total Patients

```
Total Patients = COUNT(Patients[PatientID])
```

(Optional)

```
Average Bill = DIVIDE([Total Billing], [Total Visits])
```

Part 5 — Report Creation

Page 1: Hospital Summary

Create the following visuals:

1. **Card** – Total Patients
2. **Card** – Total Visits
3. **Card** – Total Billing
4. **Column Chart**
 - Axis: Department Name
 - Values: Total Visits

5. Table Visual

- Full Name
 - Age
 - Gender
 - City
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Page 2: Doctor & Department Analysis

Create the following visuals:

1. Bar Chart

- Axis: Doctor Name
- Values: Total Billing

2. Slicer

- Department Name

3. Line Chart

- Axis: Visit Date
- Values: Total Visits